jamovi cheat sheet

This cheat sheet outlines key jamovi commands and navigation steps for statistical analysis, as taught in the PHCM9795 course

Generative AI statement

This work was developed with the assistance of NotebookLM

1. jamovi Interface & Setup

- **Download & Install:** Free at `https://www.jamovi.org/download.html`. Version 2.6.26 solid is recommended. [Section 1.8]
- **Main Views: Spreadsheet view** (accessed via **Data** tab) for data entry and changes; **Analysis view** (accessed via **Analyses** tab) for statistical results. [Section 1.9]
- **Open/Save/Export:** Click the **three-bar icon** (upper-left corner). [Section 1.10]
  - **Open Data: 3-bar icon** > **Open** > **Browse**.
  - **Save Work: 3-bar icon** > **Save** (`.omv` file format, saves data and output).
  - **Export Data: 3-bar icon** > **Export** (data in other formats like `txt`, `csv`, `xlsx`, `rds`, `dta`).
  - **Export Output: 3-bar icon** > **Export** (output as `pdf` or `html`).
  - **Copy Output** to Word Processor: **Right-click** on output and choose **Export** (plain text) or **Copy table/Copy table as HTML**.
- **Install Modules: Analyses tab** > **large + (top-right)** > **jamovi library**. Search and **INSTALL** desired module (e.g., `surveymv`, `esci`). [Sections 2.15 and 3.17]

2. Data Handling & Manipulation

- **Enter Data:** Type directly into cells in the spreadsheet view. [Section 1.9]
- **Rename Variables:** Click **Data tab** > **click column header** > **Setup**, then edit the **Variable name**, **Measure type** (e.g., Nominal, Continuous), and **Levels** (e.g., for categorical data). [Section 1.9]
- **Create New Variable: Data tab** > **click empty column** > **Setup** > **NEW COMPUTED VARIABLE**. Enter variable name and formula in the `fx` box (e.g., `weight / height^2`). [Section 3.13]
- **Set Value to Missing: Click** the cell with the erroneous value and press **Delete** or **Backspace**. Always keep an original, unedited copy of your data. [Section 1.11]
- **Weight Data** (for Summarised Categorical Data): **Data tab** > **Weights**, then define the count variable as the **weight variable**. [Section 6.7]
- **Re-order Categorical Variable Levels**: In **Data** > **Setup**, select the variable, then in the **Levels** section, click and drag levels or use the arrows to re-order (e.g., to set "Active" as the first level for a group variable). This is crucial for correct calculation of relative risks and odds ratios. [Section 6.8]

3. Descriptive Statistics (Numerical)

- **Continuous** data:
  - **Analyses** > **Exploration** > **Descriptives**. Move variable(s) to **Variables** box. [Section 1.9]
  - **Interquartile Range (IQR):** In **Statistics** section, tick **Percentiles**. [Section 1.10]
  - **Standard Error of Mean** and **Confidence Interval for Mean**: In **Statistics** section, tick **Std. error of Mean** and **Confidence interval for Mean**. [Section 3.16]
  - **Summarise by Another Variable**: Move grouping variable to **Split by** box. [Section 3.14]
- **Categorical** data:
  - **One-way Frequency Table: Analyses** > **Exploration** > **Descriptives**. Move categorical variable to **Variables** box. In **Statistics** section, tick **Frequency tables**. [Section 2.12]
  - **Two-way Frequency Table** (Cross-tabulation): **Analyses** > **Frequencies** > **Contingency Tables** > **Independent Samples**. Move variables to **Rows** and **Columns**. Request percentages in **Cells** section (e.g., **Row** or **Column** percent). [Section 2.13]
  - **Working with Summarised Data**: Enter summarised data into a new spreadsheet, including a **count** variable. In **Analyses** > **Frequencies** > **Independent Samples**, move **count** variable to **Counts** field. [Section 6.10]

4. Descriptive Statistics (Graphical)

- **Continuous** data:
  - **Density** plot: **Analyses** > **Exploration** > **Descriptives**. Move variable to **Variables** box. In **Plots** section, tick **Density**. [Section 1.10]
  - **Box** plot: **Analyses** > **Exploration** > **Descriptives**. Move variable to **Variables** box. In **Plots** section, tick **Box plot**. [Section 1.10]
  - **Scatter** plot: **Analyses** > **Exploration** > **Scatterplot**. Choose variables for **X-Axis** and **Y-Axis**. Add a fitted line by ticking **Linear** in the **Regression Line** section. [Section 8.9]
- **Categorical** data (Bar Charts):
  - **One Categorical Variable**: **Analyses** > **Exploration** > **Descriptives**. Move variable to Variables box. In Plots section, tick **Bar plot**. [Section 2.14]
  - **Two Categorical Variables** (Contingency Tables command): [Section 2.15]
    - **Clustered** (Side by side): **Analyses** > **Frequencies** > **Contingency Tables** > **Independent Samples**. Choose **Plots** > **Bar Plot**, select **Bar Type: Side by side**. Adjust X-axis as needed (e.g., columns).
    - **Stacked**: Choose **Bar Type: Stacked**.
    - **Stacked Relative Frequency**: Choose Y-axis: **Percentages within column**.
  - **Two Categorical Variables** (**surveymv** module): **Modules** > **surveymv** > **Survey Plots**. Choose Variable to plot and Grouping Variable. Set Type (Stacked bar) and Frequencies (Percentages). [Section 2.15]

5. Probability Distributions

jamovi does not have built-in functions to calculate probabilities; external applets are recommended.

- **Binomial** probabilities [Section 2.17]: https://homepage.stat.uiowa.edu/~mbognar/applets/bin.html
- **Normal** probabilities [Section 3.15]: https://homepage.stat.uiowa.edu/~mbognar/applets/normal.html

6. Confidence Intervals

- **Mean** (Individual data): **Analyses** > **Exploration** > **Descriptives.** Tick **Confidence interval for Mean.** [Section 3.16]
- **Mean** (Summarised data): **Install** `esci` module. **esci** > **Means and Medians** > **Single Group**. Select **Analyze summary data** tab, enter **Mean**, **Standard deviation**, **Sample size**. Tick **Extra details** for Standard Error. [Section 3.17]
- **Proportions**: **Frequencies** > **One Sample Proportion Tests** > **2 Outcomes | Binomial Test**. Define the count variable as the analysis variable (if individual data, use actual categorical variable). Set **Test value** (for hypothesis testing) and tick **Confidence intervals**. Note: jamovi uses Clopper-Pearson interval, not Wilson. [Section 6.17]

7. Hypothesis Testing

- **One-Sample T-test**: **Analyses** > **T-Tests** > **One Sample T-Test**. Select **dependent variable**, enter **Test value** (hypothesised mean). [Section 4.11]
- and **Confidence interval for Mean**: **Analyses** > **T-Tests** > **Independent Samples T-Test**. Move continuous variable to **Dependent variables**, grouping variable to **Grouping Variable**. Tick **Welch's** for unequal variances. Request **Mean difference** and **Confidence interval** in **Additional Statistics**. [Section 5.5]
- **Paired T-test**: **Analyses** > **T-Tests** > **Paired Samples T-Test**. Enter **paired variables** into **Paired Variables** field. Select **Mean difference** and **Confidence interval** in **Additional Statistics**. [Section 5.7]

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